

# ETX1100/5900 Business Statistics

## Week 3

### Part 1: Sampling Distribution

### Part 2: Confidence Interval Estimation

**Figure 1:** ETX5900 source PDF, Week 3, page 1

# Learning Objectives

Describe the role of statistical inference and apply inference methods to single populations.

**Textbook references:**
Berenson et al *Basic Business Statistics 5th edition,*
Chapter 6 Sections 6.1-6.2
Chapter 7, Sections 7.1-7.5.
Chapter 8, Sections 8.1-8.3

**Figure 2:** ETX5900 source PDF, Week 3, page 2

# The Basics of Sampling Distribution

- Sample statistic is only an **estimate** of the truth
- Since the sample may vary, any sample statistic is not exact and has **variation/error** around them. **The smaller the error, the greater the accuracy**.
- Assume we take data samples repeatedly, and compute sample means as the statistic for each set of sample. Then we would have the **sampling distribution** of the sample mean to portray its variability
- A sampling distribution is the probability distribution of a sample statistic.

(eg: Xbar has a sampling distribution and p also has a sampling distribution)

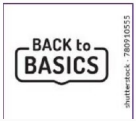



**Figure 3:** ETX5900 source PDF, Week 3, page 3

# Sampling Distribution of the Sample Mean

**The sampling distribution of the sample mean is a statistical tool that helps quantify the uncertainty of the estimate $\bar{x}$ for a given data set.**

$\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$ refers to the sampling distribution of the sample mean.

- It shows how the sample mean ($\bar{x}$ ) varies from sample to sample
- Helps us understand the properties of the sample mean as an estimator of the population mean.

If $X \sim N(\mu, \sigma^2)$, then $\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$

The variation of $\bar{x}$ is measured by the standard error of the mean- $SE(\bar{x})$.

The standard error is the standard deviation of $\bar{x}$ . It is calculated as follows: $\frac{standard\ deviation}{\sqrt{n}}$

For a **large n**, $SE(\bar{x})$ is lower, **$\bar{x}$ is more precise estimate** of the true population mean (μ)

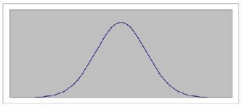
**Figure 4:** ETX5900 source PDF, Week 3, page 4

## The following are based on all possible samples of size n

$\mu_{\bar{x}} = 3.5, \sigma = 1.708$

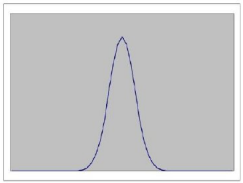
**n = 5**

$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{1.708}{\sqrt{5}} = 0.764$



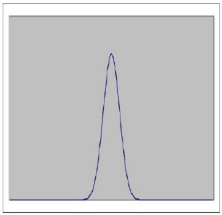
**n = 10**

$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{1.708}{\sqrt{10}} = 0.540$



**n = 25**

$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{1.708}{\sqrt{25}} = 0.342$



**Compare $\sigma_{\bar{x}}$ with $\sigma$**

Notice that $\sigma_{\bar{x}}$ is smaller than $\sigma$. The larger the sample size, the smaller $\sigma_{\bar{x}}$.

So $\bar{X}$ tends to fall closer to $\mu$ as the sample size (n) increases.

**Figure 5:** ETX5900 source PDF, Week 3, page 5

## What if the distribution of X (parent distribution) is not normal?

**It doesn't matter!**

**Statistical theory gives us a result (Central Limit Theorem or CLT) which states that:**

**If the sample size $n$ is sufficiently large, regardless of the distribution of the population,**

$$\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$$

where , $\frac{\sigma}{\sqrt{n}}$ is the standard error ($\sigma_{\bar{X}}$) - standard deviation of $\bar{X}$

**In other words, regardless of the distribution of the original population,**
**the distribution of the sample mean is approximately normal,**
**IF the sample size is large enough.**
**This approximation gets better as the sample size increases.**



**Figure 6:** ETX5900 source PDF, Week 3, page 6

# Central Limit Theorem

https://www.youtube.com/watch?v=jvoxEYmQHNM

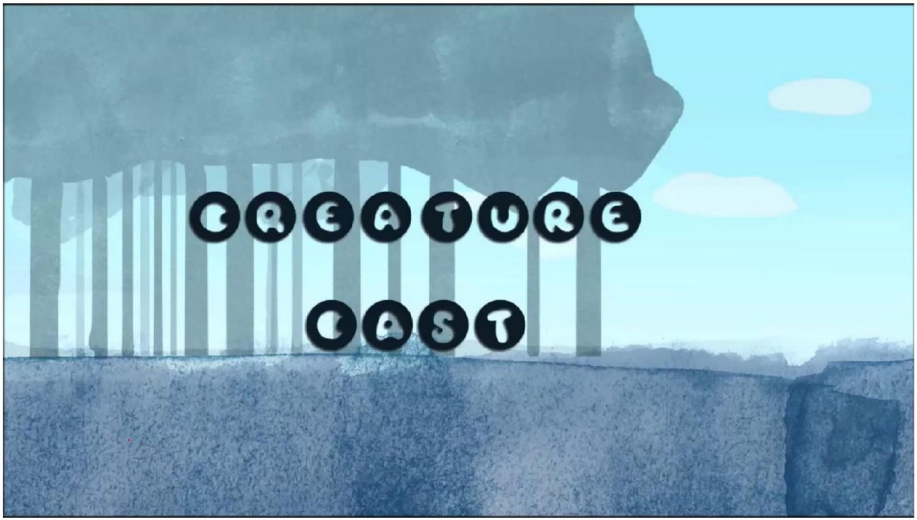



**Figure 7:** ETX5900 source PDF, Week 3, page 7

# Task 1: Customer Spending

UrbanStyle is a popular retail store that tracks customer spending to optimise staffing, stock levels, and promotional strategies. Past data indicate that the distribution of individual customer purchase amounts is normal, with mean purchase amounts of $25 and a standard deviation of $8

You have been asked to analyse spending patterns to support decisions for an upcoming promotional event.

a) The retail store defines a "moderate purchase" as spending between $15 and $35.
Using the information above, calculate the probability that a randomly selected customer makes a moderate purchase.

b) Store managers want to design a promotion that targets the top 25% of customers by spending. This will help managers set a threshold for premium rewards. Find the dollar amount such that 75% of customers spend *no more than* this value.

c) On a particular weekend, a random sample of 15 customers was taken to estimate average spending. Assuming the store's spending distribution remains unchanged, calculate the probability that the *sample mean* purchase amount for these 15 customers is not more than $30.

**Figure 8:** ETX5900 source PDF, Week 3, page 8

# Solutions for Task 1: Customer Spending

The retail store defines a "moderate purchase" as spending between $15 and $35.
Using the information above, calculate the probability that a randomly selected customer makes a moderate purchase

$$P(15 < X < 35) = P\left(\frac{15-25}{8} < \frac{X-\mu}{\sigma} < \frac{35-25}{8}\right)$$

=NORM.DIST(35,25,8,TRUE) - NORM.DIST(15,25,8,TRUE) = 0.7887

$$P(15 < X < 35) = P\left(\frac{15-25}{8} < \frac{X-\mu}{\sigma} < \frac{35-25}{8}\right)$$

$$= P(-1.25 < Z < 1.25)$$

=NORM.S.DIST(1.25,TRUE) - NORM.S.DIST(-1.25,TRUE)= 0.7887

**OR**

$=0.8944 - 0.1057$ *(from statistical tables)*

$= 0.7887$

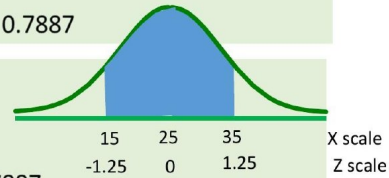



**Figure 9:** ETX5900 source PDF, Week 3, page 9

# Solutions for Task 1: Purchase Amounts

b)Store managers want to design a promotion that targets the top 25% of customers by spending. This will help managers set a threshold for premium rewards. Find the dollar amount such that 75% of customers spend *no more than* this value.

$P(X \leq x) = 0.75$

Norm.inv(0.75,25,8) = 0.303959 ≈ $30.40

OR

To get the Z value:

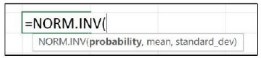
From Excel: = norm.s.inv(0.75) = 0.67449

From tables: Z=0.67 (refer to videos)

$Z = \frac{X - \mu}{\sigma}$

$0.67 = = \frac{X - 25}{8}$

X = 25 + (0.67)8 = 30.3959 ≈ $30.40



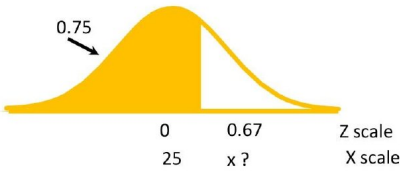




**Figure 10:** ETX5900 source PDF, Week 3, page 10

# Solutions for Task 1: Purchase Amounts

Suppose it is known that the distribution of purchase amounts by customers entering a popular retail store is normal with mean \$25 and standard deviation \$8.
Suppose a random sample of 15 customers who entered this popular retail store was taken.
c) Calculate the probability that their mean purchase amount is not more than \$30.

$$Since\ X \sim N(\mu, \sigma^2),\ \text{then}\ \bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

$$P(\bar{X} < 30) = P\left(\frac{\bar{x} - \mu}{\sigma_{\bar{x}}} < \frac{30 - 25}{\frac{8}{\sqrt{15}}}\right)$$

$$= P(Z < 2.42) = 0.9923$$

=NORM.DIST(30,25,8/SQRT(15),TRUE) = 0.9923

**Figure 11:** ETX5900 source PDF, Week 3, page 11

# Task 2: Weekly Income of Graduates

A consulting firm is preparing a report for a group of employers about graduate starting salaries. A recent national survey reports that the average weekly income of graduates one year after graduation is $600, with a standard deviation of $100. The distribution of individual weekly incomes is known to be non-normal (some graduates earn much more than others).
For a client presentation, an analyst at the consulting firm takes a random sample of 50 graduates from the survey data.

What is the probability that the mean weekly income of this sample of 50 graduates is less than $580?

**Figure 12:** ETX5900 source PDF, Week 3, page 12

# Solutions for Task 2: Weekly Income of Graduates

- Let X = weekly income of graduates one year after graduation($)
- Distn X is non-normal, so need to invoke CLT.
- Since n=50 (>30), CLT applies and states that: $\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$

$$P(\bar{X} < 580) = P\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} < \frac{580 - 600}{100/\sqrt{50}}\right)$$

P(Z<-1.414) = 0.07865

Norm.dist(580,600,100/sqrt(50),true) = 0.07865

**Figure 13:** ETX5900 source PDF, Week 3, page 13

# Task 3: Call Centre Performance Analysis (in your own time)

A telecommunications company closely monitors its call-centre operations to maintain high service standards. According to company reports, the average call handling time is 6.5 minutes, with a standard deviation of 1.2 minutes.

To verify whether daily performance aligns with expectations, a team leader reviews a random sample of 16 customer calls from the previous shift.

Assuming the company's claim is accurate and call handling times are normally distributed, the probability that the sample mean of the 16 calls falls between 6.0 and 7.0 minutes is 0.9044.

Is this probability calculation valid given the small sample size (n = 16)? Support your explanation with a discussion of **population distribution**, **sample size**, and **whether the Central Limit Theorem is required** in this case.

**Figure 14:** ETX5900 source PDF, Week 3, page 14

## Solutions for Task 3: Call Centre Performance Analysis (in your own time)

A telecommunications company closely monitors its call-centre operations to maintain high service standards. According to company reports, the average call handling time is 6.5 minutes, with a standard deviation of 1.2 minutes.
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Assuming the company's claim is accurate and call handling times are normally distributed, the probability that the sample mean of the 16 calls falls between 6.0 and 7.0 minutes is 0.9044.

Is this probability calculation valid given the small sample size (n = 16)?
Support your explanation with a discussion of **population distribution**, **sample size**, and **whether the Central Limit Theorem is required** in this case.

- Yes the answer is valid.
- Although the sample size is only 16, the parent population is normally distributed.
- We can assume that the mean of the random sample is also normally distributed.
- Therefore, CLT is not required.
- In other words, since $X \sim N(\mu, \sigma^2)$ *then* $\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$

**Figure 15:** ETX5900 source PDF, Week 3, page 15

# Part 2: Estimation of the true population mean (μ)

There are two types of estimates:

1) **Point Estimate**
   A **single** value that estimates a population ***parameter***

2) **Interval Estimate**
   A **range** of values within which the population ***parameter*** probably lies. This range is known as a confidence interval estimate

Point estimates alone do not show the uncertainty caused by sampling error.
Better approach: give a range of values within which the unknown population parameter is thought likely to lie. We refer to this range of plausible values as a **confidence interval**

**Confidence interval = plausible range of the unknown population mean given some level of confidence, based on our sample**

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**Figure 16:** ETX5900 source PDF, Week 3, page 16

# Basic Format of an Interval Estimate (of the Population Mean)

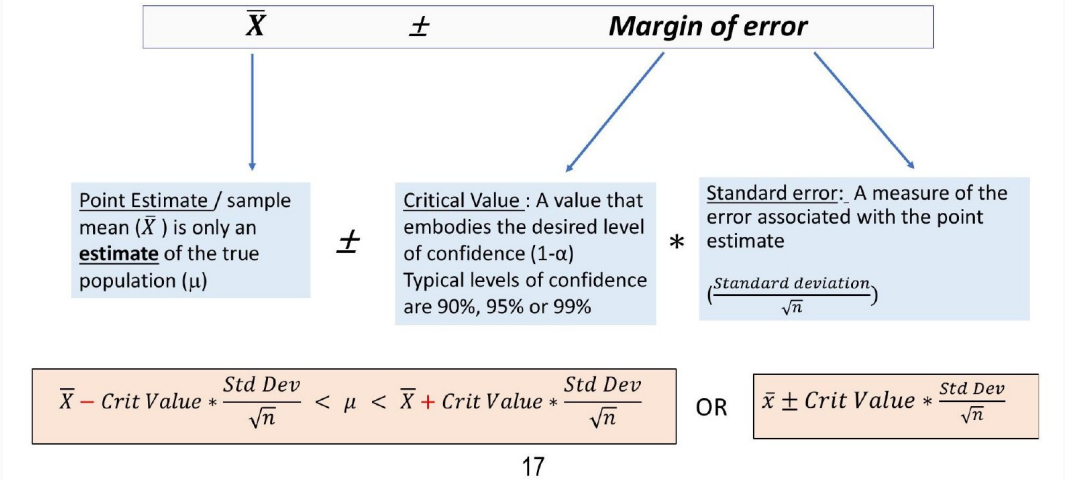



**Figure 17:** ETX5900 source PDF, Week 3, page 17

# Business Examples: Confidence Interval Estimation

1. A retail company wants to understand how much their customers typically spend per visit in their stores. They take a **random sample** of 80 customers and find that the **average spend** is **$52**, with a **standard deviation** of **$10**. Calculate a **95% confidence interval** for the average customer spend.
   - Can be used to forecast revenue, plan marketing strategies

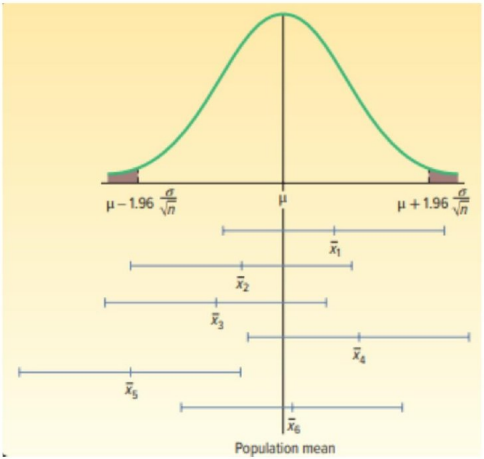
2. An investment analyst is evaluating the performance of a new mutual fund. They randomly select **36 months** of past performance data and find that the **average monthly return** is **1.8%**, with a **standard deviation** of **0.7%**. Calculate a **95% confidence interval** to estimate the fund's true average monthly return.
   - Can be used to compare with other investment options, determine if the return justifies the risk

3. A company wants to estimate the proportion of people in their target market who recognise their new logo. They randomly survey **150 people**, and **96** of them say they recognise the logo. Calculate a **95% confidence interval** to estimate the true proportion of the population that recognises the logo.
   - Can be used to guide decisions such as increasing awareness campaigns or refining their branding strategy.



**Figure 18:** ETX5900 source PDF, Week 3, page 18

# **Confidence Intervals -** in Repeated Sampling Context




E.g. 95% Confidence Level

We select a sample of $n$ observations **repeatedly** and and for **each sample** we **construct** a 95% **confidence interval** for the population mean.

We could expect 95% of intervals to contain the population mean.

While 5% of the intervals would not contain the population mean.

(Source: Lind, Marchal and Wathen, Statistical Techniques in Business Economics, 2021, 18th edition)

**Figure 19:** ETX5900 source PDF, Week 3, page 19

# Confidence Interval Estimation Formulas

$$\left(\bar{X} - Z_{\alpha/2}\frac{\sigma}{\sqrt{n}} < \mu < \bar{X} + Z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right)$$

$$\bar{X} - t_{n-1,\alpha/2}\frac{S}{\sqrt{n}} < \mu < \bar{X} + t_{n-1,\alpha/2}\frac{S}{\sqrt{n}}$$

**Estimate the true mean**

- $\mu = population\ mean$
- $\bar{X}$ = sample mean
- $Z_{\alpha/2}$ *and* $t_{n-1,\alpha/2}$ = *critical value*
- $\frac{\sigma}{\sqrt{n}}$ and $\frac{S}{\sqrt{n}}$ = std error

$$p - Z_{\alpha/2}\sqrt{\frac{p(1-p)}{n}} < \pi < p + Z_{\alpha/2}\sqrt{\frac{p(1-p)}{n}}$$

***Estimate the true proportion***

- $\pi = population\ proportion$
- $p$= *sample proportion*
- $Z_{\alpha/2}$ = *critical value*
- $\sqrt{\frac{p(1-p)}{n}}$ = *std error*

20

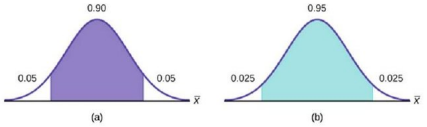
**Figure 20:** ETX5900 source PDF, Week 3, page 20

# Factors that affect the width of a Confidence Interval Estimate

If the standard deviation ($\sigma$) ↑, spread of the distribution is larger (assuming all other factors remain unchanged)
standard error $\frac{std\ deviation}{\sqrt{n}}$ ↑,
width ↑,
estimate is less precise

If the sample size (n) ↑,
standard error $\frac{std\ deviation}{\sqrt{n}}$ ↓, width ↓, estimate is more precise (assuming all other factors remain unchanged)
The bigger the sample, the more information we have to increase the precision of the interval estimate of the sample mean, the narrower the interval.

If the level of confidence (1-α) ↑,
critical value changes, width ↑ , the estimate is less precise (assuming all other factors remain unchanged)
The more confident we are, the more values we need to include in our confidence interval, the wider the interval.




$$\bar{x} \pm Z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$$

**Figure 21:** ETX5900 source PDF, Week 3, page 21

## Task 4: Estimating the Average Outstanding Balance

An audit company has been engaged to review the **outstanding customer account balances** for a mid-sized retail company. As part of the review, an analyst wants to estimate the **average (mean) outstanding balance per customer account**. A random sample of 80 accounts was selected from the company's database, finding a mean value of $86.05 and a standard deviation of $22.38.

**Calculate a 95% confidence interval estimate of the mean outstanding account balance. Show your working and interpret your result.**

**Figure 22:** ETX5900 source PDF, Week 3, page 22

## Task 4: Estimating the Average Outstanding Balance

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**Calculate a 95% confidence interval estimate of the mean outstanding account balance. Show your working and interpret your result.**

Let X = outstanding account balance ($)

Since n=80, we can invoke CLT and assume that $\bar{X} \sim N\left(\mu, \frac{\sigma^2}{80}\right)$

**Figure 23:** ETX5900 source PDF, Week 3, page 23

# Task 4: Estimating the Average Outstanding Balance

$$\bar{X} - t_{(79,0.025)} \frac{s}{\sqrt{n}} < \mu < \bar{X} + t_{(79,0.025)} \frac{s}{\sqrt{n}}$$

$Critical\ Value$: $t_{(79,0.025)=t.inv.2t(0.05,79)}$
= 1.99045

| df | 10% | 5% | 2.5% | 1% | 0.5% |
|---|---|---|---|---|---|
| 1 | 3.078 | 6.314 | 12.706 | 31.821 | 63.656 |
| 2 | 1.886 | 2.920 | 4.303 | 6.965 | 9.925 |
| 3 | 1.638 | 2.353 | 3.182 | 4.541 | 5.841 |
| 4 | 1.533 | 2.132 | 2.776 | 3.747 | 4.604 |
| 5 | 1.476 | 2.015 | 2.571 | 3.365 | 4.032 |
| 6 | 1.440 | 1.943 | 2.447 | 3.143 | 3.707 |
| 7 | 1.415 | 1.895 | 2.365 | 2.998 | 3.499 |
| 8 | 1.397 | 1.860 | 2.306 | 2.896 | 3.355 |
| 9 | 1.383 | 1.833 | 2.262 | 2.821 | 3.250 |
| 10 | 1.372 | 1.812 | 2.228 | 2.764 | 3.169 |
| 11 | 1.363 | 1.796 | 2.201 | 2.718 | 3.106 |
| 12 | 1.356 | 1.782 | 2.179 | 2.681 | 3.055 |
| 13 | 1.350 | 1.771 | 2.160 | 2.650 | 3.012 |
| 14 | 1.345 | 1.761 | 2.145 | 2.624 | 2.977 |
| 15 | 1.341 | 1.753 | 2.131 | 2.602 | 2.947 |
| 16 | 1.337 | 1.746 | 2.120 | 2.583 | 2.921 |
| 17 | 1.333 | 1.740 | 2.110 | 2.567 | 2.898 |
| 18 | 1.330 | 1.734 | 2.101 | 2.552 | 2.878 |
| 19 | 1.328 | 1.729 | 2.093 | 2.539 | 2.861 |
| 20 | 1.325 | 1.725 | 2.086 | 2.528 | 2.845 |
| 21 | 1.323 | 1.721 | 2.080 | 2.518 | 2.831 |
| 22 | 1.321 | 1.717 | 2.074 | 2.508 | 2.819 |
| 23 | 1.319 | 1.714 | 2.069 | 2.500 | 2.807 |
| 24 | 1.318 | 1.711 | 2.064 | 2.492 | 2.797 |
| 25 | 1.316 | 1.708 | 2.060 | 2.485 | 2.787 |
| 26 | 1.315 | 1.706 | 2.056 | 2.479 | 2.779 |
| 27 | 1.314 | 1.703 | 2.052 | 2.473 | 2.771 |
| 28 | 1.313 | 1.701 | 2.048 | 2.467 | 2.763 |
| 29 | 1.311 | 1.699 | 2.045 | 2.462 | 2.756 |
| 30 | 1.310 | 1.697 | 2.042 | 2.457 | 2.750 |
| 31 | 1.309 | 1.696 | 2.040 | 2.453 | 2.744 |
| 32 | 1.309 | 1.694 | 2.037 | 2.449 | 2.738 |
| 33 | 1.308 | 1.692 | 2.035 | 2.445 | 2.733 |
| 34 | 1.307 | 1.691 | 2.032 | 2.441 | 2.728 |
| 35 | 1.306 | 1.690 | 2.030 | 2.438 | 2.724 |
| 36 | 1.306 | 1.688 | 2.028 | 2.434 | 2.719 |
| 37 | 1.305 | 1.687 | 2.026 | 2.431 | 2.715 |
| 38 | 1.304 | 1.686 | 2.024 | 2.429 | 2.712 |
| 39 | 1.304 | 1.685 | 2.023 | 2.426 | 2.708 |
| 40 | 1.303 | 1.684 | 2.021 | 2.423 | 2.704 |
| 45 | 1.301 | 1.679 | 2.014 | 2.412 | 2.690 |
| 50 | 1.299 | 1.676 | 2.009 | 2.403 | 2.678 |
| 60 | 1.296 | 1.671 | 2.000 | 2.390 | 2.660 |
| 70 | 1.294 | 1.667 | 1.994 | 2.381 | 2.648 |
| 80 | 1.292 | 1.664 | 1.990 | 2.374 | 2.639 |

**Figure 24:** ETX5900 source PDF, Week 3, page 24

# Task 4: Estimating the Average Outstanding Balance

$$\bar{X} - t_{(79,0.025)} \frac{s}{\sqrt{n}} < \mu < \bar{X} + t_{(79,0.025)} \frac{s}{\sqrt{n}}$$

$$\left(86.05 - 1.990 \frac{22.38}{\sqrt{80}} < \mu < 86.05 + 1.990 \frac{22.38}{\sqrt{80}}\right)$$

$$(81.06957 < \mu < 91.03043)$$

| | |
|---|---|
| X Bar | 86.05 |
| Std deviation | 22.38 |
| Sample size | 80 |
| Std Error | 2.502160067 |
| Critical Value (t) | 1.99045021 |
| | |
| LL | 81.06957497 |
| UL | 91.03042503 |

We are 95% confident that the true mean outstanding account balance is estimated, on average, to lie between $81.07 - $91.03

**Figure 25:** ETX5900 source PDF, Week 3, page 25

## Task 5: Estimating the Average Outstanding Balance _revised (Own time)

During an audit of a company, the auditor wishes to estimate the mean outstanding account balance using a sample. It is known that standard deviation is $22.38. The sample of 80 accounts have a mean value of $86.05.

An audit company has been engaged to review the **outstanding customer account balances** for a mid-sized retail company. As part of the review, an analyst wants to estimate the **average (mean) outstanding balance per customer account**. **It is known that standard deviation is $22.38.** A random sample of 80 accounts was selected from the company's database, finding a mean value of $86.05.

**Calculate a 95% confidence interval estimate of the mean outstanding account balance. Show your working and interpret your result.**

**Figure 26:** ETX5900 source PDF, Week 3, page 26

## Task 5: Estimating the Average Outstanding Balance _revised

During an audit of a company, the auditor wishes to estimate the mean outstanding account balance using a sample. It is known that standard deviation is $22.38. The sample of 80 accounts have a mean value of $86.05.

An audit company has been engaged to review the **outstanding customer account balances** for a mid-sized retail company. As part of the review, an analyst wants to estimate the **average (mean) outstanding balance per customer account**. **It is known that standard deviation is $22.38.** A random sample of 80 accounts was selected from the company's database, finding a mean value of $86.05.

**Calculate a 95% confidence interval estimate of the mean outstanding account balance. Show your working and interpret your result.**

Let X = outstanding account balance ($)

Since n=80, we can invoke CLT and assume that $\bar{X} \sim N\left(\mu, \frac{\sigma^2}{80}\right)$

**Figure 27:** ETX5900 source PDF, Week 3, page 27

# Task 5: Estimating the Average Outstanding Balance _revised

$$\bar{X} - z\alpha/_2 \frac{\sigma}{\sqrt{n}} < \mu < \bar{X} + z\alpha/_2 \frac{\sigma}{\sqrt{n}}$$

$z\alpha/_2 = z_{0.05}/_2 = z_{0.025}$

Norm.s.inv(0.025) = -1.95996 (add a negative sign and change this to an absolute value for the purpose of substituting in the formula)

$$\left(86.05 - 1.95996\frac{22.38}{\sqrt{80}} < \mu < 86.05 + 1.990\frac{22.38}{\sqrt{80}}\right)$$

$$81.1459 < \mu < 90.9541$$

We are 95% confident that the true mean outstanding account balance is estimated, on average, to lie between $81.15 - $90.95

| | |
|---|---|
| X Bar | 86.05 |
| Std deviation | 22.38 |
| Sample size | 80 |
| Std Error | 2.50216007 |
| Critical Value (Z) | 1.95996398 |
| LL | 81.1458564 |
| UL | 90.9541436 |

**Figure 28:** ETX5900 source PDF, Week 3, page 28

## Task 6: Confidence Interval for proportions– Marketing Survey

**Refer to the data set on Moodle**

**More than half of our potential market have tried our frozen food product**

Let's use the concepts of confidence interval for proportions to validate/invalidate this claim

| Have Tried | Count of Have Tried |
|---|---|
| No | 42.17% |
| Yes | 57.83% |
| Grand Total | 100.00% |

**Figure 29:** ETX5900 source PDF, Week 3, page 29

# Task 6: Confidence Interval for proportions– Marketing Survey

**More than half of our potential market have tried our frozen food product**

Let's use the concepts of confidence interval to validate/invalidate this claim

| 95% Confidence Interval for True Proportion | |
|---|---|
| alpha | 0.05 |
| p | 0.5783 |
| SE(p) | 0.016878798 |
| Z crit | 1.968591669 |
| | |
| Lower Limit | 0.545072539 |
| Upper Limit | 0.611527461 |

We are 95% confident that the true proportion of customers who tried the product is estimated, on average, to lie between 54.52% and 61.14%

This range is well above half.

**Figure 30:** ETX5900 source PDF, Week 3, page 30

## Task 6: Marketing Survey

Let X = number of customer who have tried the product

Checks for normality:

np, n(1-p) ≥5

856*0.5783 and 856*(1-0.5783) ≥5

Since both values are greater than 5, the assumption of normality is valid.

$$\left(0.578271 - 1.959964\sqrt{\frac{0.578271 * (1 - 0.578271)}{856}} < \pi < 0.578271 - 1.959964\sqrt{\frac{0.578271 * (1 - 0.578271)}{856}}\right)$$

$$(0.578271 - 1.959964 * 0.016879 < \pi < 0.578271 + 1.959964 * 0.016879\ )$$

$$0.545189 < \pi < 0.611353$$

We are 95% confident that the true proportion of customers who tried the product is estimated, on average, to lie between 54.52% and 61.14%

**Figure 31:** ETX5900 source PDF, Week 3, page 31

## Task 7: Recruitment Service_in your own time

Suppose that, in a survey of 600 employers, 126 indicate that they have used a recruitment service within the past two months to find new staff.

a. Construct and interpret a 95% confidence interval for the population proportion of employers who have used a recruitment service within the past two months to find new staff.

b. Discuss the effect on the confidence interval estimate when you increase the level of confidence, assuming all other factors remain unchanged.

**Figure 32:** ETX5900 source PDF, Week 3, page 32

# Task 7: Recruitment Service _in your own time

Let X = number of employers who have used a recruitment service within the past 2 months to find new staff
n=600; x = 126; $\alpha$ = 0.05; p = 126/600 = 0.21

Check for normality:
np, n(1-p) ≥5
600*0.21 =126 and 600*(1-0.21) = 474
Since both values are greater than 5, the assumption of normality is valid.

$$\left(0.21 - 1.959964\sqrt{\frac{0.21 * (1 - 0.21)}{600}} < \pi < 0.21 - 1.959964\sqrt{\frac{0.21 * (1 - 0.21)}{600}}\right)$$

$$(0.177409 < \pi < 0.242591)$$

We are 95% confident that the true proportion of employers who have used a recruitment service within the past 2 months to find new staff is estimated, on average, to lie between 17.74% and 24.26%

**Figure 33:** ETX5900 source PDF, Week 3, page 33

## Task 7: Recruitment Service _in your own time

Suppose that, in a survey of 600 employers, 126 indicate that they have used a recruitment service within the past two months to find new staff.

b. Discuss the effect on the confidence interval estimate when you increase the level of confidence, assuming all other factors remain unchanged.

When the level of confidence increases, the confidence interval becomes wider.
This results in a loss of precision, which is the price you have to pay to achieve a higher level of confidence.

**Figure 34:** ETX5900 source PDF, Week 3, page 34